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Semiconductor package

Abstract

A semiconductor package includes a package substrate with a first vent hole, a first semiconductor chip mounted the package substrate, an interposer including supporters on a bottom surface of the interposer and a second vent hole, wherein the supporters contact a top surface of the first semiconductor chip, and the interposer is electrically connected to the package substrate through connection terminals. The semiconductor package further include a second semiconductor chip mounted on the interposer, and a molding layer disposed on the package substrate to cover the first semiconductor chip, the interposer, and the second semiconductor chip.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. application Ser. No. 17/539,963, filed Dec. 1, 2021, which claims priority under 35 U.S.C § 119 to Korean Patent Application No. 10-2021-0083201 filed on Jun. 25, 2021, in the Korean Intellectual Property Office, the subject matter of which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) The inventive concept relates generally to semiconductor packages, and more particularly, to semiconductor packages including at least one interposer.

(2) Contemporary and emerging electronic products increasingly demand high performance, rapid operating speeds, and compact sizes. In order to better meet these demands, the number, size and/or weight of constituent parts of such electronic products must be reduced. Accordingly, many semiconductor devices integrate a number of individual components into a single semiconductor package. In one approach the so-called “interposer” has been developed to enable vertically stacking of semiconductor chips within a semiconductor package. In this regard, the interposer has proven to be simple, inexpensive and reliable component associated with the stacking of semiconductor chips.

(3) Yet, there are a number of challenges associated with the integration of semiconductor chips and an interposer within a semiconductor package. And while electrical shorts are generally not been a problem when a flip-chip bonding techniques are used, voids may form between bumps of a semiconductor package when semiconductor chips are mounted using a flip-chip bonding technique. Accordingly, an under-fill resin layer is often formed to fill intervening space(s) between the bumps so as to prevent the occurrence of void(s).

SUMMARY

(4) Embodiments of the inventive concept provide semiconductor packages exhibiting improved structural stability and better reliability. Other embodiments of the inventive concept provide methods of fabricating semiconductor packages that improve overall yield.

(5) According to some embodiments of the inventive concept, a semiconductor package may include; a package substrate, wherein at least one first vent hole is centrally formed in the package substrate, a first semiconductor chip mounted the package substrate, an interposer including supporters on a bottom surface of the interposer and at least one second vent hole centrally formed in the interposer, wherein the supporters contact a top surface of the first semiconductor chip, and the interposer is electrically connected to the package substrate through connection terminals, a second semiconductor chip mounted on the interposer, and a molding layer disposed on the package substrate to cover the first semiconductor chip, the interposer, and the second semiconductor chip, wherein the molding layer fills the at least one first vent hole, the at least one second vent hole, a space between the first semiconductor chip and the interposer, and a space between the second semiconductor chip and the interposer.

(6) According to some embodiments of the inventive concept, a semiconductor package may include; a package substrate including a third vent hole centrally formed in the package substrate and a first vent hole laterally spaced apart from the third vent hole, a first lower semiconductor chip mounted the package substrate, a second lower semiconductor chip mounted the package substrate, an interposer including first supporters on a bottom surface of the interposer contacting a top surface of the first lower semiconductor chip, second supporters on the bottom surface of the interposer contacting a top surface of the second lower semiconductor chip, and a second vent hole centrally formed in the interposer, wherein the interposer is electrically connected to the package substrate through connection terminals, an upper semiconductor chip mounted on the interposer,

and a molding layer disposed on the package substrate to cover a combination of first lower semiconductor chip, the second lower semiconductor chip, the interposer, and the upper semiconductor chip, wherein the molding layer includes a first part filling the first vent hole, a second part filling a space between the first lower semiconductor chip and the second lower semiconductor chip and the interposer, and a third part filling a space between the upper semiconductor chip and the interposer.

(7) According to some embodiments of the inventive concept, a semiconductor package may include; a package substrate including a third vent hole centrally formed in the package substrate and a first vent hole laterally spaced apart from the third vent hole, a first lower semiconductor chip mounted the package substrate, a second lower semiconductor chip mounted the package substrate, a first interposer mounted on the package substrate, including first supporters contacting a top surface of the first lower semiconductor chip, and further including a centrally-formed second vent hole, a second interposer laterally spaced apart from the first interposer, mounted on the package substrate, including second supporters contacting a top surface of the second lower semiconductor chip, and further including another centrally-formed second vent hole, a first upper semiconductor chip mounted on the first interposer above the centrally-formed second vent hole, a second upper semiconductor chip mounted on the second interposer above the another centrally-formed second vent hole, a molding layer disposed on the package substrate to cover a combination of first lower semiconductor chip, the second lower semiconductor chip, the first interposer, the second interposer, the first upper semiconductor chip and the second upper semiconductor chip, wherein the molding layer includes a first part filling the third first vent hole and extending to cover at least a portion of a bottom surface of the package substrate, a second part filling a space between the first lower semiconductor chip and the second lower semiconductor chip and the first interposer and the second interposer, and a third part filling a space between the first upper semiconductor chip and the first interposer and the second upper semiconductor chip and the second interposer, and the first interposer is electrically connected to the package substrate through first connection terminals disposed to one side of the first lower semiconductor, and is further electrically connected to the package substrate through second connection terminals disposed to another side of the first lower semiconductor, and the second interposer is electrically connected to the package substrate through third connection terminals disposed to one side of the second lower semiconductor, and is further electrically connected to the package substrate through fourth connection terminals disposed to another side of the second lower semiconductor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1 and 2 are respective cross-sectional views illustrating semiconductor packages according to embodiments of the inventive concept.

(2) FIGS. 3, 4, 5, 6, and 7 (hereafter collectively, “FIGS. 3 to 7”) are respective plan views illustrating semiconductor packages according to embodiments of the inventive concept.

(3) FIGS. 8 and 9 are respective cross-sectional views illustrating semiconductor packages according to embodiments of the inventive concept.

(4) FIGS. 10, 11, 12, 13, 14, 15 and 16 are related cross-sectional views illustrating in one example a method of fabricating a semiconductor package according to embodiments of the inventive concept.

DETAILED DESCRIPTION

(5) Throughout the written description and drawings, like reference numbers and labels are used to denote like or similar elements and/or features. Throughout the written description certain geometric terms may be used to highlight relative relationships between elements, components

and/or features with respect to certain embodiments of the inventive concept. Those skilled in the art will recognize that such geometric terms are relative in nature, arbitrary in descriptive relationship(s) and/or directed to aspect(s) of the illustrated embodiments. Geometric terms may include, for example: height/width; vertical/horizontal; top/bottom; higher/lower; closer/farther; thicker/thinner; proximate/distant; above/below; under/over; upper/lower; center/side; surrounding; overlay/underlay.

(6) Figures (FIGS. 1 and 2 are respective cross-sectional views illustrating semiconductor packages according to embodiments of the inventive concept, and FIGS. 3, 4 and 5 are respective plan views illustrating semiconductor packages according to embodiments of the inventive concept, wherein FIG. 3 illustrates a bottom surface of a package substrate and FIGS. 4 and 5 respectively illustrate bottom surfaces of an interposer. Further in this regard, the respective cross-sectional views of FIGS. 1 and 2 may be understood as having been taken along line A-A' of FIG. 3 and/or line B-B' of FIG. 4.

(7) Referring to FIGS. 1, 2, 3, and 4, a package substrate **100** may be a printed circuit board (PCB) including conductive patterns formed on a top surface thereof. The package substrate **100** may include a core portion, an upper buildup portion disposed on a top surface of the core portion, and a lower buildup portion disposed on a bottom surface of the core portion. Here, the upper and lower build-up portions may include wiring layers associated with various electrical connections as well as vertical connection terminals extending through (e.g., penetrating) the core portion. The package substrate **100** may include alternately stacked dielectric layer(s) and wiring layer(s). The package substrate **100** may also include first substrate pads **102** and second substrate pads **104** disposed on the top surface of the package substrate **100**.

(8) For clarity of description, embodiments of the inventive concept are described in relation to an arbitrary geometric space defined in terms of a first (horizontal) direction (D1), a second (horizontal) direction D2 intersecting the first direction D1, and a third (vertical) direction D3 intersecting both the first and second directions D1 and D2. In this regard, the primary (top and bottom) surfaces of the package substrate **100** are assumed to be respective, horizontal surfaces, and the third direction D3 is assumed to be substantially orthogonal to the horizontal surfaces.

(9) Terminal pads **106** may be variously disposed on a bottom surface of the package substrate **100** to respectively connect external terminals **110**. The terminal pads **106** may be portions of internal lines of the package substrate **100** exposed on the bottom surface of the package substrate **100**. Alternately, the terminal pads **106** may be separate pads disposed on the bottom surface of the package substrate **100** and respectively connected to the internal lines of the package substrate **100**. The external terminals **110** may include solder balls or solder bumps having shape(s) and arrangement consistent (e.g.) with a ball grid array (BGA), a fine ball-grid array (FBGA), or a land grid array (LGA).

(10) In some embodiments, the package substrate **100** may include at least one first vent hole VH1 vertically penetrating the package substrate **100** (i.e., extending between the top and bottom surfaces of the package substrate **100**). The at least one first vent hole VH1 may be disposed in a substantially central portion of the package substrate **100**.

(11) In some embodiments, the at least one first vent hole VH1 may be disposed in a centrally-located, first region RG1 of the package substrate **100** substantially omitting external terminals **110** and separating (e.g., spacing apart) respective, second regions RG2. As shown in FIG. 3, the first region RG1 may extend in the second direction D2, and the second regions RG2 may be disposed in the first direction D1 on opposing sides of the first region RG1. That is, the first region RG1 may horizontally (or “laterally”) extend in the second direction D2 between the opposing second regions RG2. Thus, the at least one first vent hole VH1 may be disposed in the first region RG1 between opposing second regions RG2 in which external terminals **110** may be variously arranged.

(12) In some embodiments, the at least one first vent hole VH1 may have a width ‘w1’ (e.g., measured in the first direction D1) of about 100 μm to about 300 μm .

(13) In some embodiments, the at least one first vent hole **VH1** may include a number of first vent holes **VH1** spaced apart (e.g., in the second direction **D2**) at regular intervals along the first region **RG1**. Alternately, first vent holes **VH1** may be variously arranged in the first direction **D1** and/or the second direction **D2** (e.g., in an arrangement of columns and rows, a square lattice pattern, a grid pattern, a hexagonal lattice pattern, a honeycomb pattern, etc.). The first vent holes **VH1** may be variously sized, shaped and arranged at regular or irregularity intervals according to design.

(14) In contrast to the illustrated embodiment of FIG. 3, the first region **RG1** may have an elliptically-shape and be centrally disposed in relation to the package substrate **100**. Accordingly, a single second region **RG2** may substantially surround the elliptical first region **RG1**, and the first vent holes **VH1** may be arranged in a grid pattern or a honeycomb pattern within the first region **RG1**.

(15) Referring to FIGS. 1, 2, 3 and 4, a first semiconductor chip **200** may be disposed on the package substrate **100**. For example, the first semiconductor chip **200** may be centrally-disposed on the package substrate **100** over at least a portion of the first region **RG1**. At least one first vent hole **VH1** may be disposed below the first semiconductor chip **200**, such that at least a portion of the first semiconductor chip **200** vertically overlaps at least one of the first vent holes **VH1**. In this regard, the bottom surface of the first semiconductor chip **200** may directly oppose the top surface of the package substrate **100**.

(16) Thus, the bottom surface of the first semiconductor chip **200** may be an active surface, and the top surface of the first semiconductor chip **200** may be an inactive surface. It follows that the first semiconductor chip **200** may be disposed in a face-down when “mounted” (e.g., mechanically assembled and/or electrically connected) on the top surface of the package substrate **100**. In this regard, the first semiconductor chip **200** may be flip-chip mounted on the package substrate **100**. That is, the first semiconductor chip **200** may be electrically connected to the package substrate **100** through first chip terminals **210** (e.g., solder balls or solder bumps) disposed on the bottom surface of the first semiconductor chip **200**. Here, the term “electrically connected” denotes either direct and indirect connection (e.g., connections including intervening elements and/or material(s)). In this manner, the first chip terminals **210** may be disposed between the first substrate pads **102** of the package substrate **100** and first chip pads (not shown) on the bottom surface of the first semiconductor chip **200**.

(17) The first semiconductor chip **200** may be either a memory chip or a logic chip. Here, the memory chip may include at least one of a dynamic Random Access Memory (RAM) (DRAM), a static RAM (SRAM), a flash memory, a phase-change RAM (PRAM), a resistance RAM (ReRAM), and a magnetic RAM (MRAM), and the logic chip may be (e.g.,) an application processor (AP). However, regardless of specific type, the first semiconductor chip **200** may be electrically connected through the package substrate **100** to one or more of the external terminals **110**.

(18) An interposer substrate (hereafter, “interposer”) **300** may be mounted on the first semiconductor chip **200**. The interposer **300** may include a dielectric substrate and/or a silicon substrate, and may be vertically spaced apart from the top surface of the first semiconductor chip **200**.

(19) In some embodiments, the interposer **300** may be variously sized and/or variously mounted with respect to the first semiconductor chip **200**, such that an outer portion of the interposer **300** laterally extends beyond an outer edge of the first semiconductor chip **200**. In this regard, the interposer **300** may be understood as including an overhang region **313**, wherein third substrate pads **302** may be disposed on a bottom surface of the overhang region **313** and/or fourth substrate pads **304** may be disposed on a top surface of the overhang region **313**. Thus, in some embodiments, the interposer **300** may include a central portion (see, **RG3** of FIG. 4) bracketed (e.g., on left and/or right sides) by one or more overhang region(s) **313** (see, **RG4** of FIG. 4).

(20) However, in other embodiments, the outer edge of the interposer **300** may be vertically aligned

with the outer edge of the first semiconductor chip **200**. And in still other embodiments, the outer edge of the first semiconductor chip **200** may laterally extend beyond the outer edge of the interposer **300**.

(21) As noted above, the interposer **300** may be variously mounted on the package substrate **100**. For example, the interposer **300** may be electrically connected through connection terminals **310** to the package substrate **100**. That is, connection terminals **310** may extend between third substrate pads **302** on the overhang region **313** of the interposer **300** and second substrate pads **104** of the top surface of the package substrate **100**. Here, connection terminals **310** may be disposed on one or both side(s) of the first semiconductor chip **200**. The connection terminals **310** may include solder balls or solder bumps. For example, the connection terminals **310** may be solder balls each having a composite structure that includes a core, metallic portion and a peripheral portion surrounding the core portion.

(22) In some embodiments, the interposer **300** may additionally be wire-bonded to the package substrate **100**. For example, one or more of the third substrate pads **302** on the top surface of the interposer **300** may be wire-bonded to one or more of the second substrate pads **104** on the top surface of the package substrate **100**.

(23) A number of supporters **320** may be provided on the bottom surface of the interposer **300** (e.g., protruding from the bottom surface of the interposer **300**). Thus, the supporters **320** may vertically extend from the bottom surface of the interposer **300** to contact the top surface of the first semiconductor chip **200**, such that the supporters **320** directly support the interposer **300** on the first semiconductor chip **200**.

(24) Respective supporters **320** may include (e.g.,) a solder resist material. Respective supporters **320** may have a pillar shape (e.g., the cylindrical pillar of FIG. 4 or the polygonal pillar of FIG. 5). Respective supporters **320** may each a height 'h' (e.g., measured in the third direction D3) of about 200 μm to about 300 μm , and a width 'w2' (see, FIG. 4) of about 100 μm to about 300 μm .

(25) In some embodiments, the supporters **320** may be regularly spaced along the top surface of the first semiconductor chip **200** according to an interval 'ga' that ranges from about 500 μm to about 1,000 μm .

(26) In some embodiment, the supporters **320** may be variously arranged (e.g., in the first and second directions D1 and D2) within the a third (or central) portion RG3 of the interposer **300** that is bracketed on one or more sides by fourth region(s) RG4 in which the connection terminals **310** may be disposed. (See, e.g., FIG. 4).

(27) Thus, in some embodiments like the one illustrated in FIG. 4, the supporters **320** may be arranged in columns and/or rows, a square lattice pattern, a grid pattern, etc. Alternately, in other embodiments like the one illustrated in FIG. 5, the supporters **320** may be arranged in a hexagonal lattice pattern, a honeycomb pattern, etc. However, these are just ready examples and the scope of the inventive concept is not limited thereto.

(28) In some embodiments, a fourth regions RG4 may correspond to overhang regions **313** of the interposer **300** extending in either the first direction D1 or the second direction D2.

(29) From the foregoing, those skilled in the art will appreciate that the supporters **320** may mechanically support the interposer **300** on the first semiconductor chip **200**, thereby inhibiting or preventing warping of the interposer that may otherwise occur due to process(es) (e.g., a solder reflow process) used to mount a second semiconductor chip **400** on the top surface of the interposer **300**. Further, since the respective supporters **320** are respectively separated by defined horizontal distance(s) (e.g., 'ga'), and the interposer **300** is separated from the first semiconductor chip **200** by a defined vertical distance (e.g., 'h'), a molding layer **500** may be readily formed between the interposer **300** and the first semiconductor chip **200**. This process and its results will be further described in some additional detail hereafter.

(30) In some embodiments, the interposer **300** may further include at least one second vent hole VH2 vertically penetrating the interposer **300**. Here, the at least one second vent hole VH2 may be

centrally disposed in the interposer **300** (e.g., within the third (or central) region RG3 of the interposer **300**). That is, the at least one second vent hole VH2 may be disposed between the supporters **320** within the third region RG3.

(31) In some embodiments, respective second vent holes VH2 may have a width 'w3' of about 100 μm to about 300 μm .

(32) Referring to FIG. 4, second vent holes VH2 may be laterally spaced apart (e.g., in at least one of the first direction D1 and the second direction D2) at regular intervals. For example, assuming the connection terminals **310**—arranged in columns extending in the second direction D2 and/or rows extending in the first direction D1 within respective fourth regions RG4 of the interposer **300**—the second vent holes VH2 may be arranged in the second direction D2. In such an arrangement, the second vent holes VH2 allow molding material to easily fill space(s) between the interposer **300** and the first semiconductor chip **200** in order to form the molding layer **500**. In this regard, the term "molding material" is used to denote one or more materials constituent to the molding layer **500**.

(33) In this regard, it should be noted that disposition of various connection elements within an intended flow path of molding materials may significantly impede the flow of such molding materials, particularly in the first direction D1 towards center portions of an interposer/semiconductor chip combination. Thus, embodiments of the inventive concept further facilitate the introduction of molding materials through the second vent holes VH2 in the interposer **300**. This approach and its results will be describe in some additional detail hereafter.

(34) However, it should be further noted that no limitation is imposed on the arrangement of the second vent holes VH2 which may be variously arranged (e.g.,) in columns and/or rows, a square lattice pattern, a grid pattern, a hexagonal lattice pattern, a honeycomb pattern, etc.

(35) In some embodiments, the second vent holes VH2 may be variously disposed between and among the supporters **320**. In this regard in some embodiments, a separation interval between second vent holes VH2 may be greater than the interval 'ga' between the supporters **320**.

(36) Referring to FIGS. 1 and 2, the second semiconductor chip **400** may be centrally mounted on the interposer **300** within the third region RG3, wherein at least one of the second vent holes VH2 is disposed below the second semiconductor chip **400**. Thus, the second semiconductor chip **400** may vertically overlap at least one of the second vent holes VH2, such that at least one of the second vent holes VH2 is disposed between the first semiconductor chip **200** and the second semiconductor chip **400**.

(37) In some embodiments, the second semiconductor chip **400** may have a bottom surface directly opposing the top surface of the interposer **300**. The bottom surface of the second semiconductor chip **400** may be an active surface, and the top surface of the second semiconductor chip **400** may be an inactive surface.

(38) Accordingly, the second semiconductor chip **400** may be mounted on the top surface of the interposer **300** (e.g., using a flip-chip technique). For example, the second semiconductor chip **400** may be electrically connected to the interposer **300** through second chip terminals **410** (e.g., solder balls or solder bumps) disposed on the bottom surface of the second semiconductor chip **400**. The second chip terminals **410** may be provided between second chip pads (not shown) on the bottom surface of the second semiconductor chip **400** and the fourth substrate pads **304** on the interposer **300**.

(39) The second semiconductor chip **400** may also be a memory chip or a logic chip.

(40) With this configuration and regardless of specific chip type, the second semiconductor chip **400** may be electrically connected to the package substrate **100** through the interposer **300**. The embodiment illustrated in FIG. 1 shows a single second semiconductor chip **400**, but the scope of the inventive concept are not limited thereto and multiple second semiconductor chips **400** (e.g., vertically stacked semiconductor chips) may be mounted on the interposer **300**. Here, however, multiple second semiconductor chips **400** may be spaced apart across one or more of the second

vent holes VH2.

(41) Referring to FIG. 1, the molding layer **500** may be provided on the package substrate **100** and may include one or more molding material(s). Thus, the molding layer **500** may be provided on the top surface with the package substrate **100** to substantially cover the combination of the first semiconductor chip **200**, the interposer **300**, and the second semiconductor chip **400**. For example, the molding layer **500** may cover at least side surfaces of the first semiconductor chip **200**, side surfaces of the interposer **300**, side surfaces of the second semiconductor chip **400**, and a top surface of the second semiconductor chip **400**. In some embodiments, the first semiconductor chip **200**, the interposer **300**, and the second semiconductor chip **400** may be substantially embedded within the molding layer **500** on the top surface of the package substrate **100**.

(42) Further in this regard and referring to FIGS. 1 and 2, the molding layer **500** may substantially fill space(s) between the package substrate **100** and the first semiconductor chip **200**. Thus, the molding layer **500** may include have a first part **510** substantially covering the bottom surface of the package substrate **100** as well as filling the space between the package substrate **100** and the first semiconductor chip **200**. A portion of the first part **510** may be disposed in respective first vent holes VH1, as well as the bottom surface of the package substrate **100**. Thus, in some embodiments, the first part **510** of the molding layer **500** may extend through the first vent holes VH1 of the package substrate **100** onto the bottom surface of the package substrate **100** from the space between the package substrate **100** and the first semiconductor chip **200**.

(43) The first part **510** may substantially cover the bottom surface of the package substrate **100** to a first thickness 't1' less than a second thickness 't2' of the external terminals **110** (e.g., less than a distance between the bottom surface of the package substrate **100** and a bottom surface of the external terminal **110**). The resulting shape of the first part **510** on the bottom surface of the package substrate **100** (e.g., a planar spread) may depend on the arrangement of the first vent holes VH1. For example, if the first vent holes VH1 are arranged in the second direction D2 as shown in FIG. 3, the first part **510** may form with a linear shape largely extending in the second direction D2. Alternately, as shown in FIG. 2, the first part **510** may cover an entire bottom surface of the package substrate **100**. In this case, the first part **510** may serve as a passivation layer that protects the bottom surface of the package substrate **100**. Alternately, the first part **510** may form as one or more islands arranged (or extending) in the second direction D2 on the bottom surface of the package substrate **100**. (See. e.g., FIG. 3).

(44) Still further, the molding layer **500** may fill space(s) between the interposer **300** and the first semiconductor chip **200**, and/or space(s) between the interposer **300** and the second semiconductor chip **400**. Thus, in some embodiments, the molding layer **500** may have a second part **520** substantially filling space(s) between the interposer **300** and the first semiconductor chip **200**, and/or a third part **530** substantially filling space(s) between the interposer **300** and the second semiconductor chip **400**.

(45) The second part **520** may substantially surround the supporters **320** between the interposer **300** and the first semiconductor chip **200**, and may contact side surface(s) of the supporters **320**. The third part **530** may substantially surround the second chip terminals **410** between the interposer **300** and the second semiconductor chip **400**, and may contact with side surface(s) of the second chip terminals **410**. In this regard, the second part **520** and the third part **530** may be connected through the second vent holes VH2.

(46) Thus, in some embodiments, during fabrication of a semiconductor package, the second vent holes VH2 may serve as flow paths for molding material forming the molding layer **500**, as the molding layer **500** is introduced between the interposer **300** and the first semiconductor chip **200**, as well as between the interposer **300** and the second semiconductor chip **400**. This approach and its results will be described in some additional detail hereafter. Therefore, the molding layer **500** may fill space between the interposer **300** and the first semiconductor chip **200**, space between the interposer **300** and the second semiconductor chip **400**, and space within the second vent holes

VH2.

(47) FIGS. 6 and 7 are respective plan views illustrating bottom surfaces of semiconductor packages according to embodiments of the inventive concept.

(48) Referring to FIGS. 1, 6, and 7, the interposer 300 may again include a centrally disposed third region RG3 in which the supporters 320 may be variously arranged. However, the third region 320 may be surrounded on four (4) sides by a peripheral fourth region RG4 (e.g., extending along outer edges of the interposer 300) in which connection terminals 310 may be variously arranged.

(49) Thus, the third region RG3 may be a central portion of the interposer 300, and in some embodiments, may correspond in lateral area to that of the first semiconductor chip 200, such that the supporters 320 may be disposed in contact with the top surface of the first semiconductor chip 200.

(50) As before, the interposer 300 may include second vent holes VH2 disposed within the third region RG3 of the interposer 300, and disposed between and among the supporters 320.

(51) In the illustrated example of FIG. 6, the second vent holes VH2 are arranged in both first direction D1 and second direction D2 to form a cross pattern, as yet another example of possible arrangement patterns. Such an arrangement of the second vent holes VH2 allows molding material to easily fill space(s) between the interposer 300 and the first semiconductor chip 200 when the molding layer 500 is formed. For example, the second vent holes VH2 may be used to introduce the molding material into the third region RG3—which may be difficult to do because of the surrounding connection terminals 310. This approach and its results will be described in some additional detail hereafter.

(52) In relation to the foregoing and referring to FIG. 7, the package substrate 100 may include a cross-shaped first region RG1 in which the first vent holes VH1 are disposed, and multiple second regions RG2 separated by the cross-shaped first region in which the external terminals 110 may be disposed. Further, as shown in FIG. 7 and assuming the first vent holes VH1 are arranged in the cross-shaped pattern of FIG. 6, the first part 510 of the molding layer 500 may be formed with a cross-shaped pattern.

(53) FIG. 8 is a cross-sectional view showing a semiconductor package according to embodiments of the inventive concept.

(54) Referring to FIG. 8, multiple first semiconductor chips 200 are spaced apart and mounted on the top surface of a central portion of the package substrate 100. First vent holes VH1 may be disposed below each one of the first semiconductor chips 200. That is, each one of the first semiconductor chips 200 may vertically overlap at least one of the first vent holes VH1. However, as previously described in relation to FIG. 3, the arrangement of first vent holes VH1 may vary by design.

(55) The semiconductor package may further include at least one third vent hole VH3 vertically penetrating the package substrate 100, and disposed within the central portion of the package substrate 100. In some embodiments, each of the third vent holes VH3 may be disposed between the first semiconductor chips 200, and may have a width of about 100 μm to about 300 μm .

(56) The third vent holes VH3 may be spaced apart at regular intervals in at least one of the first direction D1 and the second direction D2.

(57) The interposer 300 may be disposed over the first semiconductor chips 200 and mounted on the package substrate 100 through the connection terminals 310. FIG. 8 shows connection terminals 310 disposed laterally outside of the first semiconductor chips 200, but the scope of the inventive concept are not limited thereto. Alternately or additionally, one or more connection terminals 310 may be disposed between the first semiconductor chips 200.

(58) The interposer 300 may again include the supporters 320 extending from the bottom surface of the interposer 300 to contact with the top surfaces of respective first semiconductor chips 200. At least one second vent hole VH2 may be disposed in the central portion of the interposer 300 between the first semiconductor chips 200.

(59) The second semiconductor chip **400** may be centrally disposed on the interposer **300** such that the second vent hole **VH2** is disposed below the second semiconductor chip **400**.

(60) The molding layer **500** may again be provided to cover at least the combination of the first semiconductor chips **200**, the interposer **300**, and the second semiconductor chip **400**.

(61) Thus, the molding layer **500** may fill space(s) between the package substrate **100** and the interposer **300**, extend from a space between the interposer **300** and the first semiconductor chips **200** through a space between the first semiconductor chips **200** into a space between the package substrate **100** and the first semiconductor chips **200**. In addition, the molding layer **500** may extend from space between the package substrate **100** and the first semiconductor chips **200** through the first and third vent holes **VH1** and **VH3** onto the bottom surface of the package substrate **100**.

(62) FIG. **9** is a cross-sectional view illustrating a semiconductor package according to embodiments of the inventive concept.

(63) Referring to FIG. **9**, multiple (e.g., first and second) lower semiconductor chips **200** may be mounted on the package substrate **100**, and multiple (e.g., first and second) interposers **300** may be respectively disposed on the first semiconductor chips **200**. Thus, in contrast to the embodiment of FIG. **8** wherein a single interposer is disposed over multiple first semiconductor chips, the embodiment of FIG. **9** includes a first lower semiconductor chip **200a** and a second lower semiconductor chip **200b** laterally spaced apart and mounted on the substrate package **100**, as well as a first interposer **300a** disposed on the first lower semiconductor chip **200a**, and a second interposer **300b** disposed on the second lower semiconductor chip **200b**. However, this is just one possible arrangement of lower semiconductor chips and interposers, and the scope of the inventive concept is not limited thereto.

(64) Here, the structure and arrangement of interposers **300a** and **300b** in relation to the lower semiconductor chips **200a** and **200b** may be substantially the same as that previously described in relation to FIGS. **1**, **2**, **3**, **4**, and **5**.

(65) The semiconductor package of FIG. **9** may include at least one fourth vent hole **VH4** vertically penetrating from top to bottom surfaces of the package substrate **100**. The at least one fourth vent hole **VH4** may be centrally provided in the package substrate **100** (e.g., between the first interposer **300a** and the second interposer **300b** and/or between the first lower semiconductor chip **200a** and the second lower semiconductor chip **200b**). The at least one fourth vent hole **VH4** may have a width of about 100 μm to about 300 μm .

(66) The at least one fourth vent hole **VH4** may include a plurality of fourth vent holes **VH4** spaced apart at regular (or irregular) intervals in at least one of the first direction **D1** and the second direction **D2**.

(67) A first upper semiconductor chip **400a** may be mounted on the first interposer **300a**, and a second upper semiconductor chip **400b** may be mounted on the second interposer **300b**.

(68) The molding layer **500** may be provided on the top surface of the package substrate **100** to substantially cover the combination of the first and second, lower semiconductor chips **200a** and **200b**, the first and second interposers **300**, and the first and second upper semiconductor chips **400a** and **400b**. In some embodiments, this combination of components may be substantially embedded within the molding layer **500**.

(69) During fabrication of semiconductor packages according to the inventive concept, molding material (ultimately forming the molding layer **500**) may be readily introduced into a space between the first and second lower semiconductor chips **200a** and **200b** and package substrate **100**, a space between the first and second interposers **300** and the first and second lower semiconductor chips **200a** and **200b**, and a space between the first and second interposers and the first and second upper semiconductor chips **400a** and **400b**, wherein this introduction of molding materials is facilitated by the provision of the first vent holes **VH1**, the second vent holes **VH2**, the third vent holes **VH3**, and the fourth vent holes **VH4**, as described above.

(70) Here, the first semiconductor chip **200a** may be laterally separated from the second

semiconductor chip **200b** by a gap extending (e.g.,) in the first horizontal direction **D1**.

(71) Further, the first interposer **300a** may be electrically connected to the package substrate **100** by a first set of connection terminals **310** disposed on one side of the first lower semiconductor chip **200a**, and may be further electrically connected to the package substrate **100** by a second set of connection terminals **310** disposed on another side of the first lower semiconductor chip **200a**. Similarly, the second interposer **300b** may be electrically connected to the package substrate **100** by a third set of connection terminals **310** disposed on one side of the second lower semiconductor chip **200b**, and may be further electrically connected to the package substrate **100** by a fourth set of connection terminals **310** disposed on another side of the second lower semiconductor chip **200b**.

(72) Therefore, within this configuration, the second set and the third set of the connection terminals may be disposed in the gap laterally separating the first lower semiconductor chip **200a** from the second lower semiconductor chip **200b**.

(73) FIGS. **10**, **11**, **12** and **13** are related cross-sectional views illustrating a method of fabricating a semiconductor package according to embodiments of the inventive concept, and FIG. **14** is an enlarged view of section 'A' indicated in FIG. **13** further illustrating flow of molding material.

(74) Referring to FIGS. **3**, **4**, and **10**, the first vent hole(s) **VH1** are formed in a central portion of the package substrate **100**. A mechanical drilling process, for example may be used to form the first vent hole(s) **VH1**. Respective first vent holes **VH1** may have a width of about 100 μm to about 300 μm . However, widths less than about 100 μm may impede the flow of molding material during subsequent processing.

(75) The first semiconductor chip **200** may be mounted on the package substrate **100** before or after the formation of the first vent hole(s) **VH1** using the approach(es) described in relation to FIGS. **1**, **2**, **3**, and **4** (e.g., using first chip terminals **210**).

(76) Referring to FIGS. **3**, **4**, and **11**, the second vent hole(s) **VH2** may be centrally formed (e.g., using a mechanical drilling process) in the interposer **300** to a width of about 100 μm to about 300 μm . Here again, widths less than 100 μm may impede the flow of molding material member during subsequent processing.

(77) The bottom surface of the interposer **300** may include the supporters **320**, wherein the supporters **320** may be formed by coating the bottom surface of interposer **300** with solder resist material(s), curing the solder resist material(s), and then patterning the cured solder resist material(s) to selectively form a desired pattern of protrusions that serve as the supporters **320**.

(78) The interposer **300** may be mounted (e.g., using connection terminal **310**) on the package substrate **100** in relation to the first semiconductor chip **200** and/or the second vent hole(s) **VH2**, as described above.

(79) Referring to FIGS. **3**, **4**, and **12**, the second semiconductor chip **400** may be mounted on the interposer **300** in relation to the second vent hole(s) **VH2**, as described above (e.g., using second chip terminals **410**).

(80) The formation of the second chip terminals **410** may involve the use of a thermocompression process, wherein potentially-damaging, downward pressure is applied to the interposer **300**. However, embodiments of the inventive concept provide a number and arrangement of supporters **320** sufficient to prevent damage to the interposer **300** and/or the first semiconductor chip **200** during the mounting of the second semiconductor chip **400**. Accordingly, due to an improved structure embodiments of the inventive concept are less likely to include defects (e.g., warpage, short circuits, broken connections, etc.) in the interposer **300**, the connection terminals **310**, the first chip terminals **210** and/or the second chip terminals **410**. This outcome increases overall production yield for the semiconductor package.

(81) Referring to FIGS. **3**, **4**, and **13**, the molding layer **500** may be formed on the top surface of the package substrate **100** to cover at least the combination of first semiconductor chip **200**, the interposer **300**, and the second semiconductor chip **400**.

(82) Further, the molding material **550** used to form the molding layer **50** may be introduced to fill

a space between the package substrate **100** and the first semiconductor chip **200**, a space between the first semiconductor chip **200** and the interposer **300**, and a space between the interposer **300** and the second semiconductor chip **400**.

(83) The enlarged view of FIG. **14** further highlights (using directional arrows) the improved introduction of molding material **550** provided by embodiments of the inventive concept. For example, a first portion of the molding material **550-1** may be introduced (e.g., injected) into the space between the interposer **300** and the second semiconductor chip **400**, and may flow through the second vent hole(s) **VH2** into the space between the interposer **300** and the first semiconductor chip **200**.

(84) Of note, the connection terminals **310** may hinder the flow of molding material **550-1** into the space between the interposer **300** and the first semiconductor chip **200**. However, the molding material **550-1** may flow from the space between the interposer **300** and the second semiconductor chip **400** through the second vent holes **VH2** and into the space between the interposer **300** and the first semiconductor chip **200**.

(85) Of further note, the presence of the supporters **320** between the interposer **300** and the first semiconductor chip **200** further facilitates the flow of molding material **550-1** between the interposer **300** and the first semiconductor chip **200**. As a result, the molding material **550-1** may readily fill the space between the interposer **300** and the first semiconductor chip **200**, as indicated by arrows in FIG. **14**.

(86) Of still further note, the flow rate of the molding material **550**, once injected, may vary by region (or location) within the semiconductor package. For example, the flow rate of the molding material **550** may be less between the interposer **300** and the first semiconductor chip **200** than between the package substrate **100** and the first semiconductor chip **200**.

(87) In some embodiments, the molding material **550** may be introduced via multiple flow paths within the package substrate **100**, thereby increasing the fill rate for the molding material **550** within respective spaces.

(88) However, regardless of molding material introduction points it is possible that one or more spaces within the semiconductor package may not be completely filled with molding material **550**. Alternately or additionally, one or voids (**VO**) may be formed within the molding material **550** during its introduction. More troubling in many comparative examples, there is no outwardly detectable indicator that the spaces have not been filled or that voids are present in the molding layer **500**.

(89) Embodiments of the inventive concept may provide a single-direction (e.g., top-down) or multi-directional flow of molding material **550**, such that a second (e.g., a vented or overflow) part of the molding material **550-2** flows (or extrudes) from the space between the package substrate **100** and the first semiconductor chip **200** into the first vent hole(s) **VH1**. Therefore, even though there is an increase in filling of the molding material **550-2** in spaces between the substrates **100** and **300** and the semiconductor chips **200** and **400**, the molding material **550-2** may be outwardly discharged through the first vent holes **VH1**, and the flow of the molding material **550-2** may be maintained in the spaces between the package substrate **100** and interposer **300** and the first and second semiconductor chips **200** and **400**. Accordingly, the second part of the molding material **550-2** may easily fill the spaces between the package substrate **100** and interposer **300** and the first and second semiconductor chips **200** and **400**, and voids **VO** may be outwardly discharged along the second part of the molding material **550-2** through the first vent holes **VH1**.

(90) Referring to FIGS. **1**, **3**, **4**, and **13**, the molding material **550** may be cured to form a molding layer **500**. The first part **510** of the molding layer **500** may fill the space between the package substrate **100** and the first semiconductor chip **200** and flow onto the bottom surface of the package substrate **100**. A second part **520** and a third part **530** of the molding layer **500** may fill the space between the interposer **300** and the first semiconductor chip **200**, as well as the space between the interposer **300** and the second semiconductor chip **400**, and the second vent holes **VH2**.

(91) Here, it should be noted that semiconductor packages including a molding layer that fails to fill spaces between various substrates and/or semiconductor chips and/or includes a surfeit of voids is vulnerable to external impacts. Further, any voids or gaps in an injected molding material may expand during a subsequent curing process and stress or fracture constituent layers and/or elements.

(92) However, embodiments of the inventive concept provide the molding layer **500** that effectively fills spaces between the various substrates, interposers and semiconductor chips, and reduces voids within molding material by discharging same through a well-positioned vent hole(s). Accordingly, semiconductor packages fabricating according to methods of the inventive concept provide a markedly lower defect rate due to improved structural stability.

(93) Referring back to FIG. **2**, the external terminals **110** disposed on the bottom surface of the package substrate **100** may be protected by the presence of the first part **510** of the molding layer **500** covering the bottom surface of the package substrate **100**. In this regard, the exposed first part **510** of the molding layer **500** may be patterned to expose the first substrate pads **102** on the bottom surface of the package substrate **100** before formation of the external terminals **110**.

(94) FIGS. **15** and **16** are respective cross-sectional views illustrating wafer-level processing of semiconductor packages according to embodiments of the inventive concept.

(95) Referring to FIG. **15**, each package substrate **100** disposed on the surface of a wafer may be understood as including a device region DR on which semiconductor packages are formed and a separation region SR allowing safe separation of adjacent semiconductor packages.

(96) Thus, a fifth vent hole VH5 may be formed in the separation region SR (e.g., during the mechanical drilling process used to form the first vent holes VH1).

(97) First semiconductor chips **200** may be mounted on respective package substrates **100** within respective device regions DR, such that each first semiconductor chips **200** is disposed over at least one of the first vent holes VH1. Afterwards, first chip terminals **210** may be used to mount the first semiconductor chips **200** on the package substrate **100**.

(98) Respective interposers **300** may be mounted on the package substrate **100** in alignment with a corresponding first semiconductor chip **200** so as to properly position supporters **320** and second vent holes VH2 in relation to the first semiconductor chips **200**. Thereafter, connection terminals **310** may be used to mount the interposers **300** on the package substrate **100**.

(99) Respective second semiconductor chips **400** may be mounted on a corresponding interposer **300**. That is, the second semiconductor chip **400** may be aligned with a corresponding interposer **300** so as to position the second semiconductor chip **400** over at least one of the second vent holes VH2. Thereafter, second chip terminals **410** may be used to mount the second semiconductor chips **400** on the interposer **300**.

(100) Referring to FIG. **16**, the molding layer **500** may be formed on the package substrates **100**. For example, a molding material may be coated over the package substrates **100** to cover the respective combinations of first semiconductor chip **200**, interposer **300**, and second semiconductor chip **400**.

(101) And as described above, the molding material may fill a space between the package substrate **100** and the first semiconductor chip **200**, a space between the first semiconductor chip **200** and the interposer **300**, and a space between the interposer **300** and the second semiconductor chip **400**. That is, the molding material may be introduced into the space between the package substrate **100** and the first semiconductor chip **200**, may flow from this space the first, second, third, fourth and/or fifth vent holes VH1, VH2, VH3, VH4 and VH5 onto the bottom surface of the package substrate **100**.

(102) Thereafter, the molding material may be cured to form the molding layer **500**.

(103) Once the protective molding layer **500** has been formed, a singulation process (e.g., a mechanical sawing process) may be applied to the wafer to separate semiconductor packages from each other, wherein each fifth vent hole VH5 may serve as an initial sawing point provided on the package substrate **100** to lessen the impact or stress on the package substrate **100** associated with

the sawing process. That is, the combination of the completed molding layer **500** according to embodiments of the inventive concept, along with the fifth vent holes **VH5** may markedly reduce the possibility of fracturing or damaging components of the respective semiconductor packages. (104) Thereafter, external terminals **110** may be added to the bottom surface of the package substrate **100** of each separated semiconductor package. (105) Semiconductor packages according to embodiments of the inventive concept may be configured such that supporters support an interposer on a semiconductor chip, thereby reducing damage (e.g., warpage) during subsequent process(es) used to mount a semiconductor chip on the interposer (e.g., solder reflow process(es)). In addition, the supporters may securely maintain a vertical separation distance between the interposer and the semiconductor chip, thereby allowing molding material to readily fill a space between the interposer and the semiconductor chip. (106) A method of fabricating semiconductor packages according to embodiments of the inventive concept provides a molding layer that readily fills spaces between various substrate(s), interposer(s) and semiconductor chip(s) by enabling an outward discharge of voids within constituent molding material. Accordingly, the defect rate of the resulting semiconductor packages may be reduced due to improved structural stability. (107) Although the inventive concept have been described in connection with some embodiments of the inventive concept illustrated in the accompanying drawings, it will be understood by one of ordinary skill in the art that variations in form and detail may be made therein without departing from the spirit and essential feature of the inventive concept. The above disclosed embodiments should thus be considered illustrative and not restrictive.

Claims

1. A semiconductor package, comprising: a package substrate; a first semiconductor chip mounted on the package substrate; an interposer including supporters on a bottom surface of the interposer and a first vent hole centrally formed in the interposer, wherein the supporters contact a top surface of the first semiconductor chip, and the interposer is electrically connected to the package substrate through connection terminals; second semiconductor chips mounted on the interposer; and a molding layer disposed on the package substrate to cover the first semiconductor chip, the interposer, and the second semiconductor chips, wherein the first vent hole is placed between the second semiconductor chips, and wherein the molding layer fills the first vent hole, a space between the first semiconductor chip and the interposer, and a space between the second semiconductor chips and the interposer.
2. The semiconductor package of claim 1, wherein the package substrate includes a second vent hole centrally formed in the package substrate, and wherein the molding layer further fills the second vent hole.
3. The semiconductor package of claim 2, wherein the second vent hole is below the first semiconductor chip, and wherein the molding layer extends through the second vent hole to cover at least a portion of a bottom surface of the package substrate.
4. The semiconductor package of claim 3, further comprising a plurality of external terminals on the bottom surface of the package substrate, wherein a distance between the bottom surface of the package substrate and a bottom surface of the molding layer is less than a distance between the bottom surface of the package substrate and bottom surfaces of the external terminals.
5. The semiconductor package of claim 1, wherein the interposer includes a central region disposed above the first semiconductor chip and an overhang region disposed to one side of the central region, wherein the first vent hole and the supporters are disposed in the central region, and wherein the connection terminals are disposed in the overhang region.
6. The semiconductor package of claim 1, wherein the interposer includes an elliptically-shaped central region disposed below the second semiconductor chips and a peripheral region surrounding

the central region, and wherein the supporters are arranged in one of a grid pattern and a honeycomb pattern.

7. The semiconductor package of claim 1, wherein each one of the supporters has a height of about 20 μm to about 300 μm .

8. The semiconductor package of claim 1, wherein, when viewed in a plan view, the first vent hole is between the supporters.

9. The semiconductor package of claim 1, further comprising a third semiconductor chip mounted on the package substrate, wherein the third semiconductor chip is horizontally spaced apart from the first semiconductor chip between the package substrate and the interposer, wherein the package substrate has a third vent hole between the first semiconductor chip and the third semiconductor chip, wherein the molding layer further fills a space between the third semiconductor chip and the interposer, and wherein the molding layer extends through the third vent hole to cover at least a portion of a bottom surface of the package substrate.

10. A semiconductor package, comprising: a package substrate including a first vent hole centrally formed in the package substrate; a first lower semiconductor chip mounted on the package substrate; a second lower semiconductor chip mounted on the package substrate; an interposer including first supporters on a bottom surface of the interposer contacting a top surface of the first lower semiconductor chip, and second supporters on the bottom surface of the interposer contacting a top surface of the second lower semiconductor chip, wherein the interposer is electrically connected to the package substrate through connection terminals; an upper semiconductor chip mounted on the interposer; and a molding layer disposed on the package substrate to cover a combination of first lower semiconductor chip, the second lower semiconductor chip, the interposer, and the upper semiconductor chip, wherein the first vent hole is placed between the first lower semiconductor chip and the second lower semiconductor chip, and wherein the molding layer fills a space inside of the first vent hole, a space between the first lower semiconductor chip and the second lower semiconductor chip and the interposer, and a space between the upper semiconductor chip and the interposer.

11. The semiconductor package of claim 10, wherein the package substrate further includes second vent holes laterally spaced apart from the first vent hole, and wherein the molding layer further fills the first vent hole and the second vent holes.

12. The semiconductor package of claim 11, wherein the second vent holes are below the first lower semiconductor chip and the second lower semiconductor chip.

13. The semiconductor package of claim 11, wherein the molding layer extends through the first vent hole and the second vent holes to cover at least a portion of a bottom surface of the package substrate.

14. The semiconductor package of claim 13, further comprising a plurality of external terminals on the bottom surface of the package substrate, wherein a distance between the bottom surface of the package substrate and a bottom surface of the molding layer is less than a distance between the bottom surface of the package substrate and bottom surfaces of the external terminals.

15. The semiconductor package of claim 11, wherein the interposer includes a third vent hole centrally formed in the interposer, and wherein the molding layer further fills the third vent hole.

16. The semiconductor package of claim 15, wherein the third vent hole is disposed between the first lower semiconductor chip and the second lower semiconductor chip, and wherein the first supporters are disposed in the first lower semiconductor chip and on the second lower semiconductor chip.

17. The semiconductor package of claim 15, wherein, when viewed in a plan view, the third vent hole is between the first supporters.

18. The semiconductor package of claim 15, wherein the upper semiconductor chip includes a plurality of upper semiconductor chips, wherein the upper semiconductor chips are horizontally spaced apart from each other, and wherein the third vent hole is disposed between the upper

semiconductor chips.

19. The semiconductor package of claim 10, wherein each one of the first and second supporters has a height of about 20 μm to about 300 μm .

20. A semiconductor package, comprising: a package substrate including a first vent hole centrally formed in the package substrate; lower semiconductor chips mounted on the package substrate and horizontally spaced apart from each other; an interposer mounted on the package substrate, including a second vent hole centrally-formed in the interposer; upper semiconductor chips mounted on the interposer and horizontally spaced apart from each other; and a molding layer disposed on the package substrate to cover a combination of lower semiconductor chips, the interposer, and the upper semiconductor chips, wherein the first vent hole is placed between the lower semiconductor chips, wherein the second vent hole is placed between the upper semiconductor chips, wherein the molding layer includes a first part filling the first vent hole and extending to cover at least a portion of a bottom surface of the package substrate, a second part filling a space between the lower semiconductor chips and the interposer, and a third part filling a space between the upper semiconductor chips and the interposer, and wherein the interposer is electrically connected to the package substrate through connection terminals disposed to one side of the lower semiconductor chips.
